

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I NEELAM NANDINI [192472277] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Nandini.

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Step 1: Initial State

No doctors are assigned.

D1 = _

D2 = _

D3 = _

Step 2: Assign D1

Try assigning Morning to D1.

D1 = Morning

Constraint Check

- D1 is not assigned Night.
- Shift is available.

Assignment is valid.

Current Assignment:

D1 = Morning

D2 = _

D3 = _

Step 3: Assign D2

Try Morning

D2 = Morning

Constraint Check:

- Morning is already assigned to D1.

This assignment is invalid.

Backtrack and try another shift.

Try Afternoon

D2 = Afternoon

Constraint Check:

- Afternoon is available.

Assignment is valid.

Current Assignment:

D1 = Morning

D2 = Afternoon

D3 = _

Step 4: Assign D3

Try Afternoon

D3 = Afternoon

Constraint Check:

- Afternoon is already assigned to D2. ✗

Invalid assignment.

Backtrack and try another shift.

Try Night

D3 = Night

Constraint Check:

- D3 is not assigned Morning.
- Night is available.
- D2 works before D3.
 - D2 = Afternoon
 - D3 = Night
 - Afternoon comes before Night.

Assignment is valid.

Current Assignment:

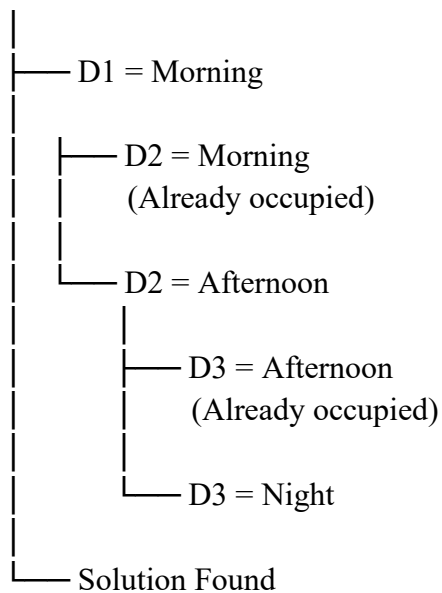
D1 = Morning

D2 = Afternoon

D3 = Night

Backtracking Search Tree

Start



Final Constraint Verification

- D1 is not assigned Night.
- D2 works before D3.
- Only one doctor is assigned to each shift.
- D3 is not assigned Morning.
- Every doctor is assigned exactly one shift.

Final Valid Schedule

Morning → D1

Afternoon → D2

Night → D3

Conclusion

Using the Backtracking Search algorithm, invalid assignments were rejected through backtracking, and a valid assignment was found by satisfying all the given constraints.

Final Answer:

- D1 → Morning
- D2 → Afternoon
- D3 → Night

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. **Movement allowed: Up, Down, Left, Right**
- ii. **Cost of each move = 1**
- iii. **Cannot pass through obstacles**
- iv. **Use Manhattan Distance heuristic**

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Given Constraints

- Move only Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles (X)
- Use Manhattan Distance heuristic

Grid 5*5

```
S . . X .  
. X . X .  
. X . . .  
. . X X .  
X . . . G
```

BFS Search Steps

Step 1: Start at S (0,0)
Queue: (0,0)

Step 2: Expand (0,0)
Add (1,0) and (0,1)
Queue: (1,0), (0,1)

Step 3: Expand (1,0)
Add (2,0)
Queue: (0,1), (2,0)

Step 4: Expand (0,1)
Add (0,2)
Queue: (2,0), (0,2)

Step 5: Continue expanding nodes until the goal is reached.

Optimal Path

S
↓
(1,0)
↓
(2,0)
↓
(3,0)
→
(3,1)
↓
(4,1)
→
(4,2)
→
(4,3)
→
G

Manhattan Distance

Formula

$$h(n) = |x_g - x| + |y_g - y|$$

At Start (0,0): $h = |4-0| + |4-0| = 8$

At Goal (4,4): $H=0$

Total Cost

- Number of moves = 8

- Cost per move = 1

Total Cost = 8

Conclusion

Using BFS, the robot explores the grid level by level, avoids obstacles, and finds the shortest path from Start (S) to Goal (G) with a minimum cost of 8.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.**
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.**
- Demonstrate how the robot applies AI concepts such as:**
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.**

Given constraints.

(a) The robot must reach the survivor (G) safely while following these constraints:

- Move only Up, Down, Left, Right.
- Cannot move through blocked cells (obstacles).
- Can observe only adjacent cells (limited sensing).
- Normal movement cost = 1.
- Risk zone cost = 1 + 2 = 3.
- Must choose the minimum-cost and safest path.
- Updates its map whenever new obstacles are found (online search).

(b)Solution Approach (Uniform Cost Search + Online Search)

Step 1: Start from S.

Step 2: Observe adjacent cells.

Step 3: Calculate movement cost.

- Normal cell = 1
- Risk zone = 3

Step 4: Choose the path with the lowest total cost.

Step 5: If a new obstacle is detected, update the map and search again.

Step 6: Continue until the robot reaches G.

(c) AI Concepts Applied

Intelligent Agent The robot senses the environment, makes decisions, and moves automatically toward the goal.

Rationality The robot always selects the safest and lowest-cost path.

Environment Type

- Partially Observable – Only nearby cells are visible.
- Dynamic – The environment may change.
- Sequential – Each move affects the next move.
- Discrete – The environment is represented as grid cells.

(d) Justification

Uniform Cost Search (UCS) with an Online Search Agent is the best choice because:

- It finds the minimum-cost path.
- It avoids risky and blocked areas.
- It can adapt when new obstacles are discovered.
- It ensures safe and efficient navigation during rescue operations.

Conclusion

The rescue robot uses Uniform Cost Search and online search to safely reach the survivor while minimizing total cost. This approach satisfies all the given constraints and is suitable for a disaster-affected environment.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided